21. $f(x) = x + \frac{3}{x}$ on $[1, 5]$.

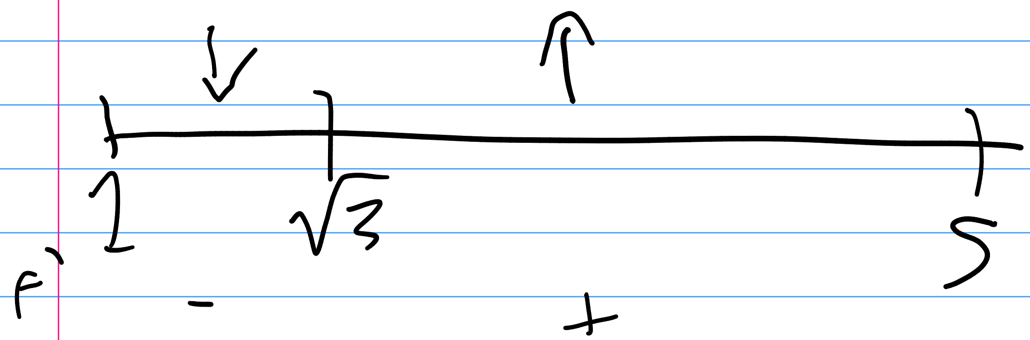
$1 + \frac{-3}{x^2}$

$0, -\sqrt{3}, \sqrt{3}$ $\quad 1, 5, \sqrt{3}$

$f(1) = 4$

$f(5) = 5.6$ ← global max

$f(\sqrt{3}) = 2\sqrt{3} \approx 3.46$ ← global min



$\int a\,dx \Rightarrow ax + C$

$-(0)x + 3$

21. $\int x\cos(x^2)\,dx$

$u = x^2$

$du = 2x\,dx$

$\frac{1}{2}\int 2x\cos(x^2)\,dx$

$\frac{1}{2}\int \cos(u)\,du$

$\frac{1}{2}(\sin(x^2) + C)$

29. $\int \frac{e^x}{e^x+1}\,dx$

$\int e^x(e^x+1)^{-1}\,dx$

$u = e^x+1$

$du = e^x\,dx$

$\int (u)^{-1}\,dx$

$\ln(u) + C$

$\ln(e^x+1) + C$

$$\int_{0}^{2} \cos(3x-1)\,dx \qquad u=3x-1$$

$$du=3$$

$$\frac{1}{3}\int_{-1}^{5} \cos(u)\,du$$

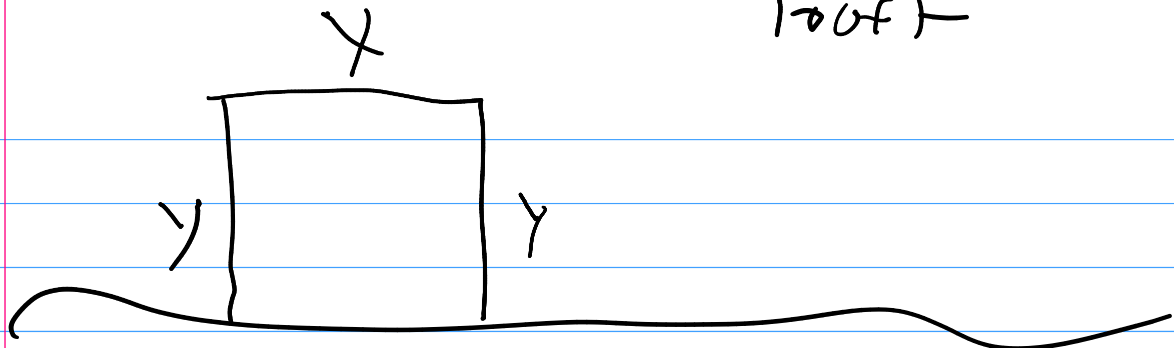
$$\frac{1}{3}\sin(u)+C \Big|_{-1}^{5}$$

$$\frac{1}{3}\left(\sin(5)-\sin(-1)\right)$$

$$\frac{1}{3}\left(\sin(3x-1)+C\right)\Big|_{0}^{2}$$

$$\frac{1}{3}\left(\sin(5)-\sin(-1)\right)$$

100ft



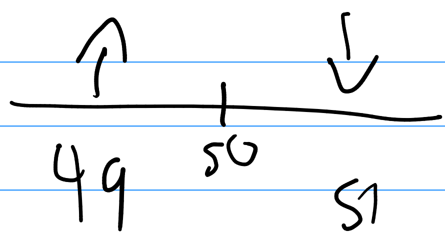
$$2y + x = 100$$

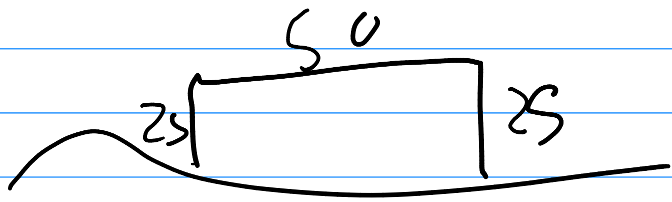
$$yx = A \qquad y = \frac{100-x}{2}$$

$$A = \left(\frac{100-x}{2}\right)x = 50x - \frac{1}{2}x^2$$

$$A' = 50 - x$$

$$x = 50$$





1250 ft^2

L'Hôpital's Rule

$$\lim_{x \to 0} \frac{\sin x}{2x}$$ co-oprnate

$$\lim_{x \to c} \frac{f(c)}{g(c)} = \frac{f'(c)}{g'(c)}$$ coöperate

Iff indeterminate

$$\lim_{x \to 1} \frac{(x^2 - 1)}{x^2 + 2x - 3} \qquad \frac{0}{0}$$

$$\frac{2x}{2x+2} \qquad \frac{2}{4}$$

$$\frac{x+1}{x+3}$$

$$\int u\,dv = uv - \int v\,du$$

$$\int x\cos x\,dx \qquad \begin{aligned} u &= x \\ dv &= \cos x\,dx \end{aligned}$$

$$x\sin x - \int \sin x\,dx$$

$$x\sin x - \cos x + C$$

$$1 + \varepsilon + 3\varepsilon^2$$

$$F_{spring} = -kx$$

$$F = ma$$

$$x''(t) = -k\,x(t)$$

$$\frac{d^2}{dx^2} \sin x \qquad \frac{d}{dx}\left(\frac{d}{dx}\right)$$

$$-\sin x \qquad \frac{d^2}{(dx)^2}$$

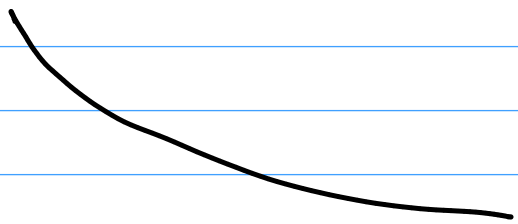
$$-A \sin x$$

$$-A \sin(bx)$$

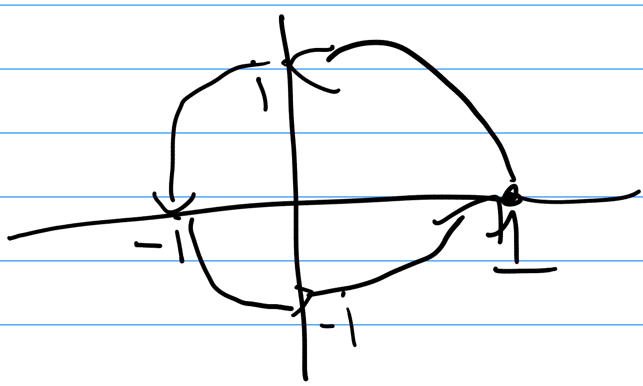
Ansatz

$$-A \sin(bx+c)$$

$$e^{ix}$$



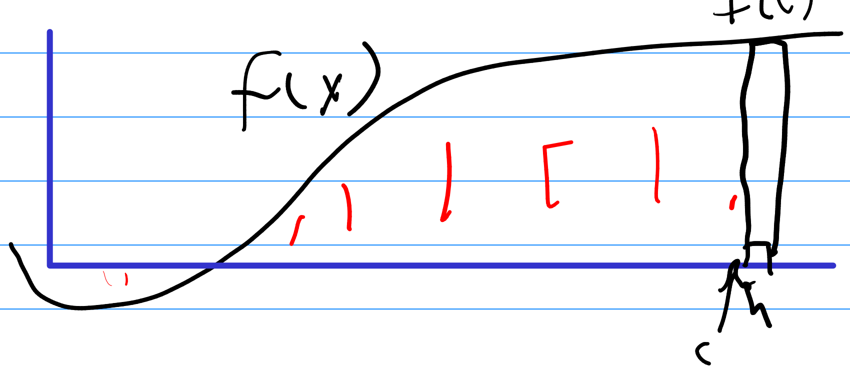
$$e^{i\pi}+1=0$$



$$Pe^{rt}$$

$$e^{ix} = \cos x + i \sin x$$

$$(x-1)(x-1)\ (x+i)$$



$$F(x)$$

$$F(x+h) \approx F(x) + f(x)h$$

$$F(x+h) - F(x) \approx f(x)\, h$$

$$f(x) \approx \frac{F(x+h) - F(x)}{h}$$

$$f(x) = \lim_{h \to 0} \frac{F(x+h) - F(x)}{h}$$

$$f(x) = F'(x)$$